

Protocol

The role of C5a in Rheumatoid and Psoriatic Arthritis in comparison with Osteoarthritis - Profiling of Synovial and Systemic Biomarkers

Lars Hornum¹, Anker Jon Hansen¹, Ditte Tornehave¹, Marianne Scheel Fjording¹, Paula Colmenero³, Inger Falbe Wätjen², Niels Henrik Sørensen⁴, Henning Bliddal^{2,5}, Else Marie Bartels²

¹Novo Nordisk A/S, Måløv, Denmark

²The Parker Institute, Copenhagen University Hospital, Bispebjerg and Frederiksberg, Frederiksberg, Denmark

³Radcliffe Department of Medicine, University of Oxford, Oxford, UK

⁴Hand Section, Department of Orthopaedics, Herlev and Gentofte Hospital, Gentofte, Denmark

⁵Faculty of Health and Medical Sciences, University of Copenhagen, Copenhagen, Denmark

Abstract

Complement activation correlates to rheumatoid arthritis disease activity, and increased amounts of the complement split product C5a is observed in synovial fluids from rheumatoid arthritis patients. Blockade of C5a or its receptor (C5aR) is efficacious in several arthritis models. The aim of this study is to investigate the role of C5a and C5aR in human rheumatoid arthritis and psoriatic arthritis – both with respect to expression and function.

Synovial fluid, blood and synovial samples will be obtained from rheumatoid arthritis, psoriatic arthritis and osteoarthritis patients as a less inflammatory arthritis type, and blood from healthy subjects. Cells infiltrating synovial tissue will analysed by immunohistochemistry and flow cytometry. SF and blood will be analysed for biomarkers by flow cytometry or ELISA. The effect of a blocking anti-human C5aR mAb on leukocyte migration will be determined using a Boyden chamber. Appropriate statistical tests will be applied for comparisons.

Introduction

The introduction of biological agents for the treatment of rheumatoid arthritis has significantly changed the course of this disabling disease. Despite this improvement, only a fraction of the patients obtains an adequate clinical response to TNF- α antagonists.¹ Moreover, the risk of cardiovascular disease is significantly elevated in patients with RA, and it remains to be seen if new biological agents can block the atherogenic process and hence improve cardiovascular co-morbidity.²⁷

The role of complement in the pathogenesis of RA has historically been described in both human and animal studies, and C5a has been hypothesised to play a pivotal role in the disease process.⁷ Elevated levels of C5a have been observed in synovial fluid samples and both chondrocytes and synovial mast cells from RA patients have been shown to express the C5a receptor.⁸⁻¹¹ In support, C5a-stimulated human neutrophils are a well-known source of various matrix metalloproteinases involved in cartilage and extracellular matrix turnover.²¹⁻²³ Moreover, C5a stimulates monocyte production of IL-8 which in RA has been shown to further activate neutrophil production of matrix metalloproteinases.^{24,25} Interestingly, both the cytokine IL-8 and Matrix Metalloprotease 3 (MMP3) has recently been shown to predict clinical outcome of infliximab and methotrexate therapy.²⁶ Despite these studies, a direct correlation between C5a activity and arthritic disease activity remains to be established in humans. A special group is the Psoriatic arthritis patients (PSA). In some way this group shows similar symptoms and biological markers as RA, but in others ways this group diverges from RA; among others, PSA shows a site-specific extra-capsular region of attack in a certain proportion of the cases³². In various joints like PIP and knee, RA and PSA have to be differentiated from osteoarthritis (OA) which may present itself with the same clinical signs and symptoms. The diseases may be separated by inflammatory markers. However, some common markers of joint inflammation cannot differ between the diseases³³.

Complement activation, indicated by the elevation of C5a, has been suggested associated with increased cardiovascular risk in patients with advanced atherosclerosis^{28,29}. Apart from being a strong chemotactic factor for atherogenic inflammatory subsets, C5a stimulates secretion of important inflammatory mediators such as TNF- α as well as the thrombolytic inhibitor PAI-1 from human mononuclear cells.^{30,31} Importantly, the C5a-mediated effects on atherosclerotic plaque macrophages are independent of TNF- α indicating a more upstream pathogenic role for C5a, when compared to TNF- α in atherosclerosis.³¹ In addition, C5a induces the expression of endothelial adhesion molecules such as P-selectin¹⁵. Both TNF- α , PAI-1, and P-selectin are important risk factors for cardiovascular disease. We hypothesize that the suggested upstream atherogenic role for C5a, when compared to TNF- α may be important in C5aR versus TNF- α -targeted prevention of cardiovascular co-morbidity in RA.

In order to tailor treatment towards patients who are most likely to benefit from targeted biological treatment, both pharmaco-genomic and -metabonomic studies have been initiated.² Several smaller studies point towards that genomic and metabonomic profiling are feasible tools in separating responders from non-responders (Lauridsen et al., unpublished results).^{3,4} In addition, important understanding of mechanism of action of novel compounds in clinical testing may be unravelled and further exploited to tailor treatment towards specific patient populations.⁵ In addition, a histological approach has shown to be predictive of long-term response to various anti-rheumatic agents, including biological agents.⁶ However, this technique requires strict adherence to specific protocols for synovial tissue sampling that is difficult to

adapt in routine clinical practice and large multi-centre clinical trials. Hence, novel biomarkers of disease activity linked to biological mechanism of action of novel compounds are needed in order to guide future treatment and dosing in rheumatoid arthritis.

Aims

The aim of the study is to evaluate the role of C5a in rheumatoid and psoriatic arthritis in comparison with osteoarthritis, and to identify biomarkers that can be used for monitoring a successful blockade of C5a signalling in the clinic.

A secondary aim is to evaluate the role of the TL1A/DR3 ligand/receptor pair in the same setting.

Objectives

Primary Objective

To evaluate the role of C5a in rheumatoid and psoriatic arthritis (RA and PsA) when compared to osteoarthritis (OA).

Secondary Objective

To identify C5a regulated genes, proteins and metabolites that can be utilized as biomarkers for successful C5aR blockade in the clinic.

Hypotheses

1. C5a is elevated in SF and plasma in RA and/or PsA patients relative to OA
2. C5a levels correlate to disease activity as defined by peripheral blood CRP-level or inflammatory cytokine levels in SF (IL-6 and TNF α)¹
3. C5a levels correlate to presence/titer of autoantibodies in plasma (RF and ACPA)
4. C5a levels in SF and plasma correlate
5. C5aR density is inversely correlated to disease activity and/or to C5a levels in SF
6. C5aR blockade will significantly inhibit chemotaxis of leukocytes (specifically DCs)

Materials and Methods

Participants

¹ Also valid for PSA: Gisondo and Girolomoni, 2009; Vandooren et al, 2009; Canete et al, 2009

A total of 30 patients with active RA, 10 patients with active PSA, and 10 patients with active OA will be included in the study.

30 healthy age- and sex-matched women and men will have blood taken for comparison with the patient groups.

Inclusion criteria patients

- Informed consent
- A diagnosis of RA according to the American College of Rheumatology (ACR1987 criteria), PSA (defined according to CASPAR (Taylor et al, Arthritis and Rheum 2006, 54 (8): 2665-73) or OA (ACR-criteria).
- Active arthritis with joint swelling and tenderness
- Patients must be fasting from midnight prior to sampling, only water intake is allowed.

Exclusion criteria patients

- An alcohol consumption of more than an amount equivalent to 1 unit of alcohol 12 hours or less prior to sampling is not allowed
- Active medical treatment for other chronic diseases than rheumatoid arthritis

Prior to study enrolment, patients must not be in active treatment with any of the following medications:

Glucocorticoid unless given in stable doses equivalent to ≤ 10 mg of prednisone orally /day the preceding month

Intra-articular, i.m. or i.v. corticosteroids within the last month

Cyclosporine

Azathioprine

Penicillamine

Etanercept

Infliximab

Abatacept

Efalizumab

Alefacept

Adalimumab

Intravenous immunoglobulins (IVIG)

Rituximab

Gold therapy

During the course of the disease, it is acceptable that the patients have been in active treatment with any of the above listed medications. In such cases, the treatment must have been discontinued at least 1 month prior to enrolment in this study.

Visits

All patients will come to the clinic for one visit. Prior to participation in any study-related activities, the subject must give written consent to participate in the study.

The following procedures are followed:

Obtain informed consent

Assess in- and exclusion criteria

Assess DAS28

Collect blood samples

Collect synovial samples

Collect urine samples

Table 1 Study population/Sample material

Hypothesis or Experimental procedure	RA	OA	PsA	SF	Plasma	Serum	Urine	Full blood
	N	N	N	µl	µl	µl	µl	µl
A1 C5a vs. OA	30	10	10	5× 20 ²	5× 20 ¹			
A2	30		10	A1 + B2 ³	A1 + B2 ⁴			

² C5a desArg ELISA

³ TNFα and IL-6 is included in RBM panel

⁴ CRP included in RBM panel

C5a vs. activity								
A3 C5a vs. auto-Ab	30			A1	A1			
A4 C5a: SF vs. plasma	30	10	10	A1	A1			
A5 C5aR	30	10	10	500,000 cells				1000
A6 chemotaxis	30	10	10	500				
B	30	10	10	5×20		5×20		

Furthermore blood will be samples for standard clinical and biochemical tests (see below).

Table 2 Synovial fluid sampling

Purpose	Volume	Procurement	Procedure	Storage “label”
C5a-desArg (A1)+ RBM + chemotaxis	1200 µl	Harvest in EDTA tubes Spin 1200 g for 10 min.	Snap-freeze supernatant in 5×20 µl + 2× 250 µl + 2×250 µl + 5×20 µl aliquots	- 80° “SF-C5a” + “SF-RBM” + “SF- chemotax”
FACS	≥ 500,000 cells	Harvest in EDTA tubes Put on ice until FACS	FACS	no

4 ml SF is harvested in EDTA tubes where possible, and – if more can be taken – 500-2000 µl is harvested in dry tubes.

Table 3 Blood Samples

Purpose	Volume	Procurement	Procedure	Storage "label"
FACS	1000 µl	Collect in EDTA tubes.	FACS	No
C5a-desArg	100 µl	Collect in EDTA tubes. Spin 1200 g for 10 min.	Snap-freeze supernatant in 20 µl aliquots	- 80° "C5a"
RBM -biomarkers	500 µl	Collect in EDTA tubes. Spin 1200 g for 10 min.	Snap-freeze supernatant in 250 µl aliquots	- 80° "RBM"

A total of 3 ml blood will be drawn into EDTA tubes. For FACS analysis 1 ml will be stored cold (+4°C) until processing for flow cytometry (to be performed at the Parker Institute). The remainder is centrifuged at 1200 g for 10 minutes. And supernatants and pellet are aliquoted as described.

Clinical and Biochemical Information

For each patient biochemical test for HDL, LDL, triglyceride concentration, total cholesterol, serum rheumatic factor, anti-CCP, CRP, erythrocyte sedimentation rate, and urine creatinine must be available. In addition information regarding smoking habits and intake of over-the-counter drugs must be made available through a questionnaire. Age, duration of disease and gender must also be available.

Flow cytometric Analyses

Flow cytometric analyses of synovial fluid cells or cells from whole blood will be performed using the BD Biosciences FACSArray instrument, which measures emitted fluorescence light in four channels only, not five channels. This makes it necessary to aliquot cells and use fluorochrome conjugated antibodies in the way stated below. A labelling of the predominant cell types in synovial fluid, namely macrophage-like synoviocytes (type A synoviocytes) and fibroblast-like synoviocytes (type B synoviocytes) is added:

FACS protocol

1. From each patient 2 x 3 FACS staining tubes (Axygen MCT-150-L-C) are set up (3 from SF and 3 from blood).
2. To 300 µl EDTA-anticoagulated whole blood are added 30 µl of a 100 µg/ml solution of human IgG (Sigma I-4506) in Staining Buffer (PBS pH 7.40 with 0.2% human serum albumin (Sigma A-1653) and 0.09% w/v sodium azide). Mixing, then standing at room temperature for 15 minutes.

3. To 400 µl of the isolated SF-cells, containing approximately 500,000 cells, are added 40 µl of a 100 µg/ml solution of human IgG in Staining Buffer (the same as used for blood samples in the above section). Mixing, then standing at room temperature for 15 minutes.

4. The now Fc-blocked whole blood is aliquoted into three 1,5 ml microcentrifuge tubes, 110 µl in each. The Fc-blocked synovial fluid cells are aliquoted into four 1,5 ml microcentrifuge tubes, 90 µl in each of these four tubes, and the rest remaining in the tube, which is then labelled "Tube 8" and is later used for staining with isotype for CD68 .

5. Add the following fluorochrome-conjugated antibodies into each set of whole blood cell tubes. The stated volumes are for staining of 1×10^6 cells, in 100 µl whole blood, but may be subject to alterations when titration for optimization has been done:

Tube 1 isotypes: 20 µl Isotype IgG1, κ -PE (BioLegend 400114)

20 µl Isotype IgG1, κ -APC (BioLegend 400122)

20 µl Isotype IgG1, κ -PE-Cy7 (BioLegend 400126)

20 µl Isotype IgG1, κ -APC-Cy7 (BioLegend 400128)

Tube 2 monocytes: 20 µl C5aR-PE (CD88) (BD 550494, Clone D53-1473, ms IgG1, κ)

20 µl CD16-APC-Cy7 (BioLegend 302018, Clone 3G8, ms IgG1, κ)

20 µl CD11b-APC (BioLegend 301310, Clone ICRF44, ms IgG1, κ)

20 µl CD14-PE-Cy7 (BioLegend 325618, Clone HCD14, ms IgG1, κ)

Tube 3 neutrophils: 20 µl C5aR-PE (CD88) (BD 550494, Clone D53-1473, ms IgG1, κ)

5 µl CD11b-APC-Cy7 (BD 557754, Clone ICRF44, ms IgG1, κ)

20 µl CD62L-PE-Cy7 (BioLegend 304822, Clone DREG-56, ms IgG1, κ)

10 µl CD203c-APC (Miltenyi 130-092-344, Clone FR3-16A11, ms

IgG1, κ)

6. Mix carefully and incubate 40 minutes at 4°C.

8. Resuspend the cells with 200 ml BD FACS Lysing Solution (BD 349202), mix, and incubate at room temperature for 10 minutes.

10. Add the following fluorochrome-conjugated antibodies into the set of synovial fluid cell tubes. The volumes for staining of 150,000 synovial fluid cells are stated, but are subject to alterations when titration for optimization has been done:

3µl Isotype IgG1, -APC-Cy7 (BioLegend 400128)

3µl CD14-PE-Cy7 (BioLegend 325618, Clone HCD14, ms IgG1,)

3 μ l CD90-APC (BD 559869, Clone SE-10, ms IgG1,[®])

10 μ l CD68-PE (BD kit 556078, Clone Y1/82A, ms IgG2b,[®]).

Tube 8 isotype control for macrophage-like synoviocytes. Cells must be permeabilized before staining:

10 µl anti- ms IgG2b, [?] (BD kit 55607)

11. Permeabilization of cells in Tubes 7 and 8 before staining with antibodies to CD68 or the isotype to this is performed after tube 7 has been stained with anti-CD88. Permeabilization is done by addition of 100 µl PBS with 1% formaldehyde, and standing for 10 minutes at 4 °C. Then addition of 600 µl Staining Buffer and 40 µl saponine solution 0.5% in PBS, then standing for 5 minutes at 4 °C. The tubes are centrifuged at RCF 300 x g for 8 minutes. Remove the supernatant, resuspend the cells in 100 µl Staining Buffer, and add 10 µl PE-anti-CD68 to tube 7 and 10 µl PE-anti-IgG 2b, [?] isotype control to Tube 8.

12. To all tubes applies that they are Vortex mixed carefully after the addition of antibodies and then incubated for 40 minutes at 4°C.

13. Wash the cells two times with Staining Buffer, 1.0 ml each time. The cells must be carefully mixed after addition of the Staining Buffer, and centrifuged at RCF 300 x g for 8 minutes. The supernatant is removed.

14. Resuspend the cells with 200 µl BD FACS Lysing Solution (BD 349202), mix, and incubate at room temperature for 10 minutes.

15. Wash the cells two times with Staining Buffer, 1.0 ml each time. The cells must be carefully mixed after addition of the Staining Buffer, and centrifuged at RCF 300 x g for 8 minutes. After the supernatant has been removed after the second wash, re-suspend cells in 150 µl Staining Buffer and transfer the cells from one microcentrifuge tube to one well in a round-bottom polypropylene microtiter plate (Nunc 267245).

16. Check the FACSArray flow cytometer with the "SPHERO Rainbow Calibration Particles (8 Peaks)" (BD 559123).

17. Set photomultiplier gains in channels FSC, SSC, Yellow, Red, Far Red and NIR appropriately and then perform the spill-over compensation using "BD Comp Beads" (BD 51-90-9001291) and the isotype control antibodies used in Tube 1. The settings, which are calculated by the instrument after the spill-over compensation samples have been run, are then saved as a template, which can also be used for later samples in this project.

18. Count the samples from Tubes 1 to 8, with injection volume 100 µl, 2.0 µl per second.

Immunohistochemistry

Human tissue samples from RA, PsA and OA, as well as from non-inflammatory controls will be studied.

Sectioning

Fixed paraffin embedded cell and tissue sections are cut at a nominal thickness of 3 µm, mounted on Superfrost® Plus microscope slides, and stored at -20°C until further processing.

Immunostaining for microscopy

The sections will be deparaffinised in xylene and rehydrated in decreasing concentrations of alcohols. Antigen retrieval will be performed in Tris-EGTA buffer (10mM; 0.5 mM), pH 9.0 in a microwave oven for 15 min. Endogenous peroxidase activity is blocked with 3% H₂O₂. Endogenous biotin is blocked by incubation with Avidin and Biotin blocking solutions for 10 min, according to the manufacturer. Non-specific binding is blocked by incubation with TBS containing 3% skimmed milk, 7% donkey serum, 3% human serum, and 3.2 mg/ml Poly-L-Lysine (PLL) for 30 min. The primary and secondary antibodies will be diluted in a Tris buffer containing 0.5% skimmed milk, 7% donkey, 3% human sera, and incubation is performed overnight at 4°C (primary antibody), and 60 min at room temperature (secondary antibody). The first amplification step will be performed by incubation with Vectastain ABC peroxidase kit, diluted in 0.1 M Tris-HCl buffer (pH 7.5) containing 0.5% Du Pont Blocking Reagent (TNB) for 30 min, followed by a second amplification step with incubation in biotinylated Tyramide for 6 min. The final amplification is performed by an additional incubation with the Vectastain ABC peroxidase kit, diluted as previously described for 30 min. The chromogenic reaction is achieved with diaminobenzidine. Nuclei will be counterstained with haematoxylin and the sections be rehydrated, cleared in xylene and mounted with Eukitt. Evaluation of the slides will be performed blinded to the observer.

Double-immunofluorescence

The immunohistochemical demonstration of the C5aR will be performed as described above. Upon the second amplification with biotinylated Tyramide for 3 min sections are incubated with the Vectastain ABC peroxidase kit, diluted as previously described for 30 min. The final amplification is performed by incubation with Alexa-594 conjugated Tyramide diluted according to manufacturer for 6 min. Sections are then subjected to second rounds of microwave oven treatment in Tris-EGTA buffer (10mM; 0.5 mM), pH 9.0 for 10 min, blocking of endogenous peroxidase activity with 3% H₂O₂, and endogenous biotin by incubation with Avidin and Biotin blocking solutions for 10 min, respectively, according to the manufacturer. Additional blocking of non-specific binding is performed by incubation with TBS containing 3% skimmed milk, 7% donkey serum, and 3% human serum for 30 min. The second primary and secondary antibodies are diluted in a Tris buffer containing 0.5% skimmed milk, 7% donkey and 3% human sera, and incubation is performed overnight at 4°C, and 60 min at room temperature, respectively. The final amplifications are performed by an additional incubation with the Vectastain ABC peroxidase kit, diluted as previously described for 30 min, and incubation with Alexa-488 conjugated Tyramide. Nuclei will be counterstained with Hoechst for 15 min, and the sections be mounted in fluorescence mounting media (Dako).

Table A1 List of antibodies and conjugated detection kits used for immunohistochemical studies

Antibodies	Vendor	Cat. #	Clone	Iso-type
Primary				
Mouse monoclonal anti-hC5aR	R&D System	MAB3648	347214	IgG2a

Antibodies	Vendor	Cat. #	Clone	Iso-type
Mouse monoclonal anti-hCD68	Dako	M0814	KP1	IgG1
Rabbit monoclonal anti-hCD3	Neomarkers	RM-9107S1	SP7	IgG
Rabbit polyclonal anti-MPO	Osenses	DOM00001G	Polyclonal	IgG
Mouse isotype specific control	R&D System	MAB002	11711	IgG1
Mouse isotype specific control	Dako	X0943	DAK-GO5	IgG2a
Rabbit monoclonal isotype specific control	Cell Signalling Technology	3900	DA1E	IgG
Rabbit polyclonal control antibody	Jackson ImmunoResearch	011-000-003	Polyclonal	IgG
Secondary				
Biotin conjugated Donkey anti-mouse	Jackson ImmunoResearch	715-065-150	Polyclonal	IgG
Biotin conjugated Donkey anti-rabbit	Jackson ImmunoResearch	711-065-152	Polyclonal	IgG
Detection systems				
Peroxidase conjugated Avidin Biotin Complex (Vectastain)	Vector	PK-6100	N/A	N/A
Biotin conjugated Tyramide	Perkin Elmer	NEL700	N/A	N/A
Alexa-488 conjugated Tyramide	Invitrogen	T-20932	N/A	N/A
Alexa-594 conjugated Tyramide	Invitrogen	T-20935	N/A	N/A

Abbreviations: MPO, Myeloperoxidase; N/A, not applicable.

Microscopic analyses

Sections are evaluated by virtual microscopy using either a Hamamatsu NanoZoomer (Hamamatsu Denmark, Ballerup, Denmark), or a FV10i Laser scanning microscope (Olympus Denmark A/S; Ballerup, Denmark), and an Olympus BX51 reflected fluorescence system microscope equipped with selective AMCA, FITC and Texas-Red filters, and a DP70 digital camera (Olympus Denmark A/S; Ballerup, Denmark).

Flow cytometry analysis of RA synovial tissue cells

Tissue

RA synovial membranes will be obtained from patients undergoing joint replacement surgery or synovectomy.

Cell Preparation for flow cytometry from synovial tissue

Single-cell suspensions of RA synovial tissue biopsies will be prepared by mechanical disruption followed by enzymatic digestion for 1-2 h using Liberase TL (Roche, Welwyn Garden City, UK) and DNase (Sigma-Aldrich Ltd, Dorset, UK). After passing the cells through a nylon mesh to exclude debris, cells are washed and resuspended in RPMI-1640 medium (containing 10% (v/v) FBS) at a density of 1×10^6 cells/ml. For FACS analysis, dead cells are excluded with Live/Dead Fixable Aqua Dead Cell Stain (Life Technologies, Waltham, MA, USA). After gating on viable CD45⁺ cells within the monocytic gate, CD14 vs. CD15 staining was used to identify macrophages and infiltrating neutrophils. The following anti-human monoclonal antibodies from BD Biosciences (San Jose, CA, USA) will be used: CD14-APC (clone M5E2), CD15-PECy7 (clone HI98), CD45-PerCP (clone TU116). Anti-human C5aR/CD88-PE (clone S5/1, Biolegend, London, UK) and mlgG2a-PE (clone MOPC-173, Biolegend, London, UK) will be used to assess C5aR expression. To block non-specific Fc receptor-mediated antibody binding, FcR Blocking reagent (Miltenyi Biotec, Lund, Sweden) is used prior to antibody staining. Stained cells are fixed in Fixation Buffer (BD Biosciences, San Jose, CA, USA). Samples will be acquired in a BD Fortessa flow cytometer (BD Biosciences, San Jose, CA, USA) and analyzed using FlowJo software (Tree Star, Ashland, OR, USA).

Ethics

Ethics Committee for London Riverside and for the Capital Region of Denmark and the Danish Data Protection Agency will be sought for permissions to carry out the study and handle the data.

Statistics

Student's t-test will be used for group comparisons, Spearman's Rank correlation test for correlations tests. To correct for multiple comparisons, Bonferroni correction will be applied.

References

1. Hetland ML, Lindegaard HM, Hansen A, Podenphant J, Unkerskov J, Ringsdal VS *et al.* Do changes in prescription practice in patients with rheumatoid arthritis treated with biological agents affect treatment response and adherence to therapy? Results from the nationwide Danish DANBIO Registry. *Ann.Rheum.Dis.* 2008;67:1023-6.
2. Ranganathan P. An update on pharmacogenomics in rheumatoid arthritis with a focus on TNF-blocking agents. *Curr.Opin.Mol.Ther.* 2008;10:562-7.

3. Lequerre T, Gauthier-Jauneau AC, Bansard C, Derambure C, Hiron M, Vittecoq O *et al.* Gene profiling in white blood cells predicts infliximab responsiveness in rheumatoid arthritis. *Arthritis Res.Ther.* 2006;8:R105.
4. Koczan D, Drynda S, Hecker M, Drynda A, Guthke R, Kekow J *et al.* Molecular discrimination of responders and nonresponders to anti-TNF alpha therapy in rheumatoid arthritis by etanercept. *Arthritis Res.Ther.* 2008;10:R50.
5. Buch MH, Boyle DL, Rosengren S, Saleem B, Reece RJ, Rhodes LA *et al.* Mode of action of abatacept in rheumatoid arthritis patients having failed TNF blockade: a histological, gene expression and dynamic MRI pilot study. *Ann.Rheum.Dis.* 2008.
6. Gerlag DM, Tak PP. Novel approaches for the treatment of rheumatoid arthritis: lessons from the evaluation of synovial biomarkers in clinical trials. *Best.Pract.Res.Clin.Rheumatol.* 2008;22:311-23.
7. Low JM, Moore TL. A role for the complement system in rheumatoid arthritis. *Curr Pharm.Des* 2005;11:655-70.
8. Jose PJ, Moss IK, Maini RN, Williams TJ. Measurement of the chemotactic complement fragment C5a in rheumatoid synovial fluids by radioimmunoassay: role of C5a in the acute inflammatory phase. *Ann.Rheum.Dis.* 1990;49:747-52.
9. Hogasen K, Mollnes TE, Harboe M, Gotze O, Hammer HB, Oppermann M. Terminal complement pathway activation and low lysis inhibitors in rheumatoid arthritis synovial fluid. *J Rheumatol* 1995;22:24-8.
10. Kiener HP, Baghestanian M, Dominkus M, Walchshofer S, Ghannadan M, Willheim M *et al.* Expression of the C5a receptor (CD88) on synovial mast cells in patients with rheumatoid arthritis. *Arthritis Rheum.* 1998;41:233-45.
11. Onuma H, Masuko-Hongo K, Yuan G, Sakata M, Nakamura H, Kato T *et al.* Expression of the anaphylatoxin receptor C5aR (CD88) by human articular chondrocytes. *Rheumatol Int.* 2002;22:52-5.
12. Goldberg BS, Weston WL, Kohler PF, Harris MB, Humbert JR. Transcutaneous leukocyte migration in vivo: cellular kinetics, platelet and C5a dependent activity. *J Invest Dermatol* 1979;72:248-52.
13. Jose PJ, Forrest MJ, Williams TJ. Human C5a des Arg increases vascular permeability. *J Immunol* 1981;127:2376-80.
14. Chenoweth DE, Hugli TE. Demonstration of specific C5a receptor on intact human polymorphonuclear leukocytes. *Proc.Natl.Acad.Sci.U.S.A* 1978;75:3943-7.
15. Foreman KE, Vaporciyan AA, Bonish BK, Jones ML, Johnson KJ, Glovsky MM *et al.* C5a-induced expression of P-selectin in endothelial cells. *J.Clin.Invest* 1994;94:1147-55.
16. Foreman KE, Glovsky MM, Warner RL, Horvath SJ, Ward PA. Comparative effect of C3a and C5a on adhesion molecule expression on neutrophils and endothelial cells. *Inflammation* 1996;20:1-9.
17. Jagels MA, Daffern PJ, Hugli TE. C3a and C5a enhance granulocyte adhesion to endothelial and epithelial cell monolayers: epithelial and endothelial priming is required for C3a-induced eosinophil adhesion. *Immunopharmacology* 2000;46:209-22.
18. Castellheim A, Pharo A, Fung M, Saugstad OD, Mollnes TE. Complement C5a is a key mediator of meconium-induced neutrophil activation. *Pediatr.Res.* 2005;57:242-7.

19. Kuroki M, O'Flaherty JT. Extracellular signal-regulated protein kinase (ERK)-dependent and ERK-independent pathways target STAT3 on serine-727 in human neutrophils stimulated by chemotactic factors and cytokines. *Biochem.J.* 1999;341 (Pt 3):691-6.
20. Monk PN, Scola AM, Madala P, Fairlie DP. Function, structure and therapeutic potential of complement C5a receptors. *Br.J.Pharmacol.* 2007;152:429-48.
21. Romanelli R, Mancini S, Laschinger C, Overall CM, Sodek J, McCulloch CA. Activation of neutrophil collagenase in periodontitis. *Infect.Immun.* 1999;67:2319-26.
22. Takafuji S, Ishida A, Miyakuni Y, Nakagawa T. Matrix metalloproteinase-9 release from human leukocytes. *J.Investig.Allergol.Clin.Immunol.* 2003;13:50-5.
23. Emingil G, Kuula H, Sorsa T, Atila G. Gingival crevicular fluid matrix metalloproteinase-25 and -26 levels in periodontal disease. *J.Periodontol.* 2006;77:664-71.
24. Van den Steen PE, Proost P, Grillet B, Brand DD, Kang AH, Van DJ *et al.* Cleavage of denatured natural collagen type II by neutrophil gelatinase B reveals enzyme specificity, post-translational modifications in the substrate, and the formation of remnant epitopes in rheumatoid arthritis. *FASEB J.* 2002;16:379-89.
25. Kuhns DB, Priel DA, Gallin JI. Induction of Human Monocyte Interleukin (IL)-8 by Fibrinogen through the Toll-Like Receptor Pathway. *Inflammation* 2007;30:178-88.
26. Visvanathan S, Marini JC, Smolen JS, Clair EW, Pritchard C, Shergy W *et al.* Changes in biomarkers of inflammation and bone turnover and associations with clinical efficacy following infliximab plus methotrexate therapy in patients with early rheumatoid arthritis. *J.Rheumatol.* 2007;34:1465-74.
27. Kaplan MJ. Is rheumatoid arthritis a risk factor for cardiovascular disease? *Nat Clin.Pract.Rheumatol* 2007.
28. Speidl WS, Exner M, Amighi J, Kastl SP, Zorn G, Maurer G *et al.* Complement component C5a predicts future cardiovascular events in patients with advanced atherosclerosis. *Eur.Heart J* 2005;26:2294-9.
29. Oksjoki R, Laine P, Helske S, Vehmaan-Kreula P, Mayranpaa MI, Gasque P *et al.* Receptors for the anaphylatoxins C3a and C5a are expressed in human atherosclerotic coronary plaques. *Atherosclerosis* 2007.
30. Okusawa S, Yancey KB, van der Meer JW, Endres S, Lonnemann G, Hefter K *et al.* C5a stimulates secretion of tumor necrosis factor from human mononuclear cells in vitro. Comparison with secretion of interleukin 1 beta and interleukin 1 alpha. *J.Exp.Med.* 1988;168:443-8.
31. Kastl SP, Speidl WS, Kaun C, Rega G, Assadian A, Weiss TW *et al.* The complement component C5a induces the expression of plasminogen activator inhibitor-1 in human macrophages via NF-kappaB activation. *J.Thromb.Haemost.* 2006;4:1790-7.
32. Marzo-Ortega H, Tanner SF, Rhodes LA, Tan AL, Conaghan PG, Hensor EM *et al.* Magnetic resonance imaging in the assessment of metacarpophalangeal joint disease in early psoriatic and rheumatoid arthritis. *Scand. J. Rheumatol.* 2009; 38(2):79-83.
33. Huang K, Wu LD. YKL-40: a potential biomarker for osteoarthritis. *J. Int. Med. Res.* 2009; 37(1):18-24.